

The Effects of Texas' Red Light Camera Ban on Driving Behavior and Police Ticketing

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Abstract

I examine the impact of a ban of red light traffic cameras in Texas after June 2019 on police officers' discretion in ticketing and driver behavior by race. I use a difference-in-differences approach to compare officer-distributed red light tickets in Austin Police Department districts (sectors) with red light traffic cameras to officer-distributed red light tickets in sectors without cameras before and after the removal of cameras. I find suggestive evidence that overall and by race, drivers are ticketed less in sectors that had a red light camera compared to sectors that did not. Results for Black drivers are significant, however there is no significant difference in estimates between races. This paper is in current progress of maintaining more data to help with analyzing effects and therefore serves as a work in progress.

1. Introduction

Every year in the United States, over 46,000 people die from car crashes. Those who are seriously injured from vehicle crashes make up 4.4 million of the population. An action taken by cities and or states to increase traffic safety was to implement a form of traffic surveillance that would influence drivers to drive safely. A response to this safety measure was the first implementation of a red light camera in New York City in 1993. These automated enforcement tools were used as a way to punish drivers with a fee for running an intersection during a red light. In theory, this would serve as a positive punishment enactment, punishing drivers with a fine to prevent them from running red lights in the future, while also generating revenue for the city.

While these cameras could improve public safety, there are also arguments against their use. For example, the cameras may cause more harm than good such as causing more crashes than preventing them. This can happen by drivers slamming on their breaks to avoid getting a red light camera ticket, but causing someone to hit them from behind (Short and Kirkpatrick, 2020). Another justification is that surveillance of vehicles may invade the privacy of drivers. There are 8 states in the U.S. who banned the use of red light traffic cameras (NCLS, 2022).

In addition to public safety and revenue generation arguments, traffic camera enforcement may have positive distributional consequences. For example, cameras only capture license plates, so the likelihood of discrimination in ticketing should be little to none. With the ban of traffic cameras, officers will have to take on more responsibility with traffic ticketing which may lead to bias in ticketing. Similarly, a ban can increase the likelihood of the use of unnecessary force if a traffic violation stop goes wrong.

Drivers may also adapt their driving behavior after the ban of red light cameras. On one hand, their tendency to run red lights may increase in areas with cameras, once they know the cameras are no longer in effect. However, this change in behavior could increase their likelihood of getting caught in running red lights by officers. On the other hand, drivers may act as if the cameras are still there, which may not change their driving behavior initially. Furthermore, their tendency to run red lights might even decrease due to the possible increase in likelihood of being caught by an officer. These concerns influence the questions of this paper: how does the ban of red light cameras affect police discretion of ticket distribution? Also, how does it impact driving behavior?

Related literature regarding red light cameras have studied the impact of implementation on various outcomes. Some papers have shown that the introduction of red light cameras decreased the violation rate at all camera and non camera intersections (Retting et al, 1999), and that overall the use of traffic cameras had a positive effect on crash reduction (Martinez-Ruiz et al, 2019). Conover et al (2022) discuss a camera system in India that monitors officers distributing traffic violations and found that tickets issued increased as the intensity of monitoring increased. These articles show that monitoring influences behavior whether the monitoring is geared towards drivers or officers. As part of my contribution, I will be looking at the removal of traffic violation enforcement, red light cameras, instead of the implementation of these cameras. A paper that also studies the removal of red light cameras, Gallagher and Fisher (2020), finds no evidence that the removal of cameras caused a reduction in the total number of accidents or injuries for Houston, Texas. An article that considers police ticketing in comparison to ticketing from speeding traffic cameras shows that officers ticket Black drivers 16 percentage points more (Xu et al, 2024). My contribution to this literature is studying the effects of a red light camera ban on police and driver behavior.

In order to measure these effects, I will first consider traffic citations distributed from both red light cameras and officers in Austin, Texas before the ban to analyze driving tendencies by race. To measure the effects of the ban of red light cameras, I then analyze the distribution of red light tickets from officers before and after the ban. Using a difference-in-differences model, I analyze the change in distribution of tickets from officers by race by comparing outcomes of tickets distributed at Austin Police Department districts (sectors) with red light cameras to tickets distributed in sectors without cameras, before and after the removal of cameras. To further test officer bias and driver tendency, I include an interaction term that tests whether there was a difference in tickets distributed in the day, where officers tend to have more discretion due to visibility from daylight, to tickets distributed at night, where officers discretion is hindered by the lack of visibility. This process is known as the “veil-of-darkness” approach. My assumption here is that officer bias during the day time can be measured against the night time since the lack of visibility composes a much smaller bias than visibility during the day. I find a reduction in the number of officer-distributed red light tickets overall and by race. There was a significant decline of 5.8 for Black drivers in sectors with cameras versus sectors without cameras. Though insignificant, results declined more for Hispanic (6.9) and White (10.5) drivers respectively. I look further to see if there is a difference in ticketing at these sectors in the day versus night time and find insignificant differences that are small overall and for Black drivers. However, the magnitude of the estimates show officers may be ticketing Hispanic drivers more at night and White drivers more during the day than at night.. At the same time, this may also show the Black drivers are more risk averse during the day time compared to Hispanic and White drivers.

2. Background

Initially, red light cameras were installed at intersections to prevent drivers from running red lights. Specifically, they were installed in intersections that Austin Police Department suggested had a high level of crash incidents (Red Light Camera Program, 2011). Their ultimate goal was to promote safety by preventing accidents (Fletcher Law Office, 2024). If a driver ran a red light, the camera would capture the license plate, and mail a citation to the owner of the vehicle. However, on June 1, 2019, prohibition of the use of red light cameras (photographic traffic signal enforcement systems) went into effect for the state of Texas. Justification for the ban explains that cameras caused more accidents instead of preventing them. For instance, a person trying to avoid a red light ticket might slam on their brakes in hopes of not running the red light which could cause the car behind to rear-end them (Short and Kirkpatrick, 2020). Another instance is a driver speeding up during a yellow light, runs the intersection during a red light (by missing the yellow light), and having an accident with oncoming traffic. Though this new law bans these traffic cameras statewide, some cities are still in contract with photo enforcement agencies, which allows them to still enforce such cameras¹.

3. Data

In order to identify red light citations distributed before and after the ban, I retrieved traffic citation data from the City of Austin Municipal Court. The dataset contains the following information on traffic stops: a description of the offense charge, the time and date of the offense, the location of the defense, the race and gender of the defendant, the agency of the officer who distributed the ticket, the officer code, and an indicator of whether the case was closed or not. I have information for both 2019 and 2020 fiscal year which provides data on citations given between October 2018 and September 2020. The bandwidth of my data will be from October 1, 2018 and February 29, 2020, which contains data eight months before and after the ban. I have restricted the data to end in February because of the COVID-19 lockdown taking place in March 2020 in Austin, Texas. I have also restricted the data to only contain traffic citations for running red lights. Thus the initial data contains 13,689 violations of running a red light. I then aggregate the number of tickets at the time, location, and race levels. Austin Police Department districts, known as sectors, provide geographic information on the location of its sectors which I use to determine the sector of each officer ticket in my sample.

The constructed data include citations from both red light cameras and officer distributed tickets. Information such as gender and race of the driver is reported for officer distributed tickets. However, since cameras only capture the license plate of the driver, such information is not

¹ The cities of Humble, Leone Valley, and Balcones Heights are still under contract.

recorded in the data for camera distributed tickets. Observations from cameras do however record the name of the driver. Therefore, I utilize the prevalence rates of race and ethnicity by first and last name as observed in the CJARS data to estimate the race of the driver.

There were ten intersections in Austin that contained red light cameras. Pre-ban, red light cameras distributed 9,928 citations, consisting of 84.5% of any red light citation received before June 1, 2019. After the ban, officer-distributed citations for running a red light made up 100% of all red light citations since cameras were no longer enforced. Table 1 shows the number of tickets distributed by red light cameras and officers by race and Figure 2 shows a visual representation by percentage. In comparison to red light cameras before the ban, there is a 13% difference in ticketing Black drivers, where officers are ticketing Black drivers slightly less, however they are ticketing Hispanic drivers more, a 60% difference, and White drivers about less, a 26% difference. After the ban, the share of officer-distributed red light tickets slightly decreased by 3% for Black drivers, increased by 14% for Hispanic drivers, and decreased by 9% for White drivers.

4. Conceptual Framework

There are a few assumptions made before analyzing the effects of the red light camera ban. For one, it is assumed that red light cameras are not biased because they only capture the license plate of the offending vehicle. This is shown in the data in which gender and race are not captured for citations distributed by the red light cameras. Another assumption is that officers will distribute more red light tickets once the cameras are uninstalled because officers have to take on more burden of distributing tickets. This can be shown by the increase in both percentage and number of red light citations handed out from officers after the ban, as shown in Table 1.

5. Empirical Design

In order to understand how the red light ticket ban impacted different racial groups, I will run a difference in difference model. This analysis will compare officer-distributed red light tickets at sectors with cameras, treated group, to officer-distributed red light tickets at sectors without cameras before and after the ban, control group. The treatment is an exogenous factor that banned red light cameras statewide on June 1, 2019. My main equation is listed below,

$$Y_{mz} = \beta_0 + \beta_1 Treated + \beta_2 Post_m + \beta_3 Treated * Post_m + X\beta_4 + \tau_m + \varepsilon_{mz} \quad (1)$$

where m is the month from the ban in which the ticket was issued and z is the closest previous red light camera region in which a red light ticket was distributed by an officer, referred to as a red light zone. The treated variable indicates if a red light ticket was distributed by an officer during the day time (Treatment = 1) versus night (Treatment = 0). The post dummy variable indicates if the citation was issued before the ban (Post = 0) or after the ban (Post = 1). The vector X controls for characteristics such as race and a seasonal time fixed effects (monthly), τ , to account for heterogeneity. The outcome variable, Y , measures the change in number of red light ticket citations handed by officers amongst race, specifically, Black, Hispanic, and White drivers. The error term captured unobserved influences.

I further test if there is a difference in officer-distributed red light tickets in camera versus non camera sectors during the daytime versus night time, before and after the ban. This process is known as a ‘veil-of-darkness’ approach. With this approach, I am able to analyze officers’ discretion during visible hours of the day to their discretion at night, when visibility is hindered, creating a secondary level of officer bias to study. This will not only inform the tendencies in officers’ discretion between different mediums, but also inform us on the capacities in which drivers’ tendencies of running red lights vary.

My identifying assumption is that if red light cameras were not to be banned, then the amount of officer-distributed red light tickets, overall and by race, would trend the same in sectors containing red light cameras compared to non red light camera sectors. I show support for the “parallel trends” assumption of similar levels in ticketing in the daytime versus nighttime for each race before the June 2019 ban of red light cameras in Figure 3.

6. Limitations

In this section, I discuss limitations that hinder the validity of my research design. The main limitation is data. Specifically, not enough information is provided to eliminate all possible avenues of endogeneity. There is not enough information provided on the officers handing out tickets such as demographics and key identifiers. Not knowing much about the officers makes it difficult to understand if results are impacted because more, or less, officers are being distributed to certain districts after the ban. Another limitation is constructing the best counterfactual. While I do believe comparing tickets in locations with and without red light cameras in police districts is a good treatment and comparison group, these areas are not mutually exclusive and hence drivers can drive through both kinds of sectors or one in particular. Lastly, a big limitation is power. As I analyze effects by race, I lose power size, which hinders the true effect. With these limitations, I take extreme caution in interpreting complete causality of results.

7. Results

I show my main results in Table 2. Using a difference-in-differences approach, I find that officers distribute red light tickets less overall and by race, handing out 5.8, 6.9, and 10.5 less tickets in sectors with cameras compared to sectors without cameras after the ban to Black, Hispanic, and White drivers respectively. However, only the estimate on Black drivers is significant.

I also consider aggregating months that were further away from June 2019, the month in which the ban occurred, on either end, before and after. I aggregated months between October 2018 and March 2019 and months between October 2019 and February 2020 together, emphasizing two months prior and after the ban individually. In Table 3, these estimates also show a decline in tickets overall and for each race after the ban, with a bigger, insignificant, estimate for White drivers (-38.7).

I further test results by running a triple interaction between indicators of camera sectors, post period, and day to show significant differences in estimates between races. Though these results are not significantly different, their signs suggest Black and Hispanic drivers are ticketed less in the day time, whereas White drivers are ticketed more in the day time in sectors with cameras. Suggestive evidence could also imply that Black and Hispanic drivers may be changing their driving behavior by being more careful in running red lights in the daytime compared to White drivers being more risk averse in the nighttime. However, a robust justification on driving behavior in this setting is difficult to access.

8. Robustness Checks

My original sample consists of officer distributed red light tickets between October 2018 and February 2020, with the ban occurring in June 2019. Since the ban occurred on the first day of June, the post period consists of nine months while the pre period consists of eight. I test whether my results are driven by the extra month in the post period. Results from dropping February 2020 are shown in Table A1 and are consistent with the main results, showing that all races receive fewer tickets in the sectors with cameras compared to sectors without cameras after the ban. Although there is a significant decline of 6.9 tickets for Black drivers, a triple interaction, comparing significant differences between races confirms that results are not significantly different by race.

9. Conclusion

Measuring bias in officer discretion as well as measuring tendencies in driver's behavior is an important avenue to study. I utilize a policy change, abolishing red light traffic cameras in the state of Texas. With these traffic cameras distributing the bulk of red light tickets, there may exist a new burden on officers in ticketing drivers for running red lights. This burden can lead to a change in bias, which could affect different drivers such as by race. Driver behavior can react differently to cameras not being in use thereafter. My results suggest the small increase in officer ticketing after the ban did not have a large significant effect on their ticket distribution. On a smaller scale, I find that all drivers received less tickets in the sectors with cameras than sectors without cameras.

This paper is in working progress with finding enough data to strengthen the identification strategy and hence bring clarity to the interpretation of results. However, this first pass can serve as a consideration to the policy change in banning red lights in Texas. As my initial analysis shows, there is a change in officer bias in red light ticketing after the ban. If the red light traffic cameras served as an unbiased form of ticketing as it only captured the license plate of the driver running the red light, and the camera's ability to handle a higher capacity of ticket distribution, then this may be more efficient than relying solely on officer distributed red light tickets. Namely, there appears to be an inequitable bias in red light ticketing for Hispanic drivers by officers, shown in Figure 2. This group may benefit the most from a reimplementation in red light traffic cameras.

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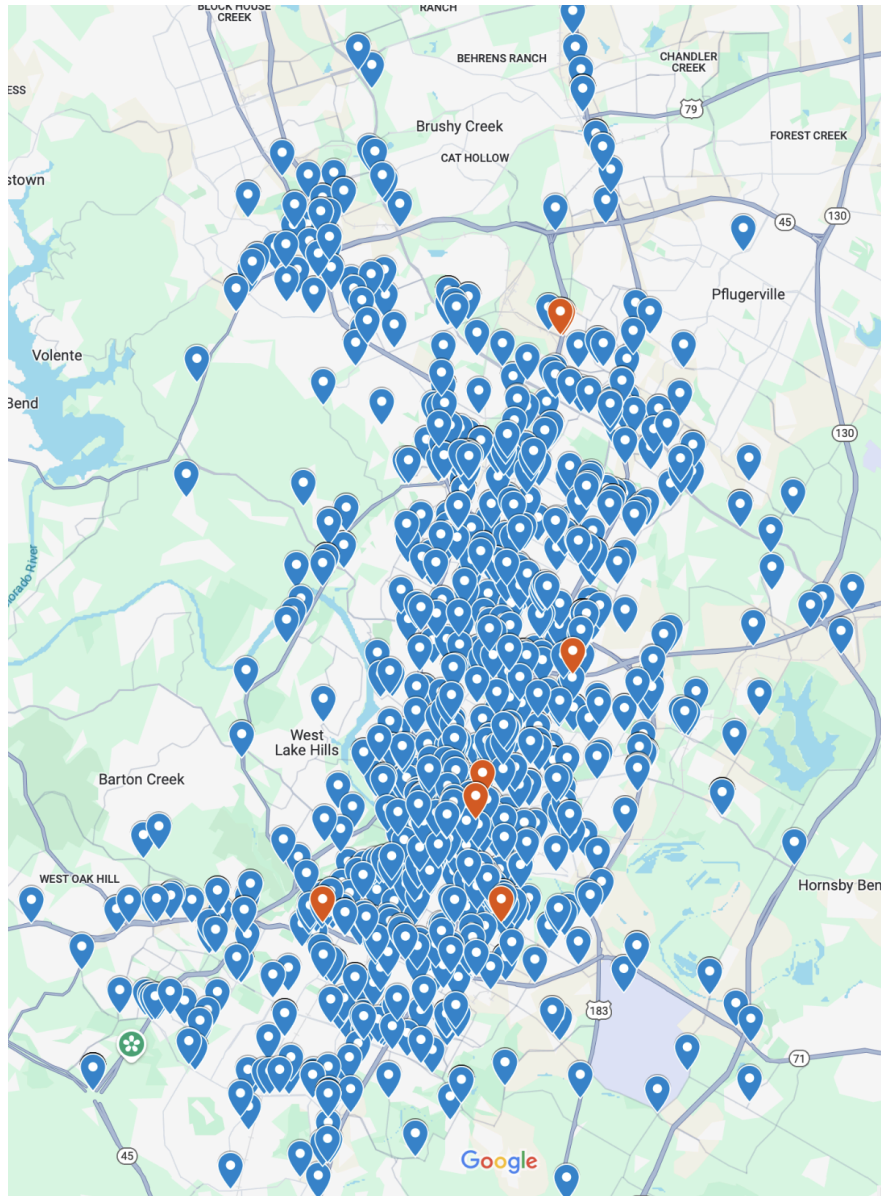
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Tables and Figures

Figure 1. Locations of distributed tickets from running a red light in Austin, TX



Note: The location of tickets distributed by Red Light Cameras are in red. There are a total of ten Red Light Cameras where some intersections consist of two cameras, one in each opposite direction of traffic. The location of tickets distributed by Officers are in blue. These tickets are dispersed throughout the city of Austin and Greater Austin, its surrounding areas included in the metropolitan area. This map is provided by Google Maps.

Figure 2. Distribution of tickets by Red Light Cameras and Officers

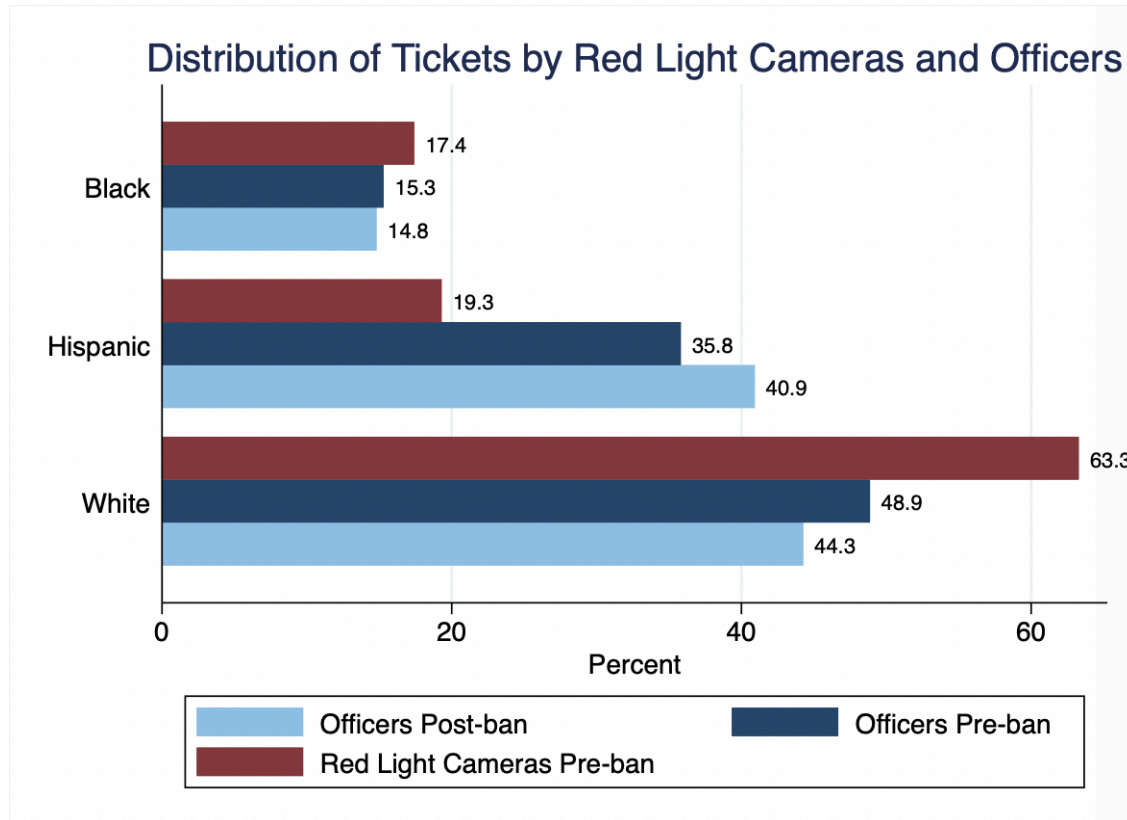
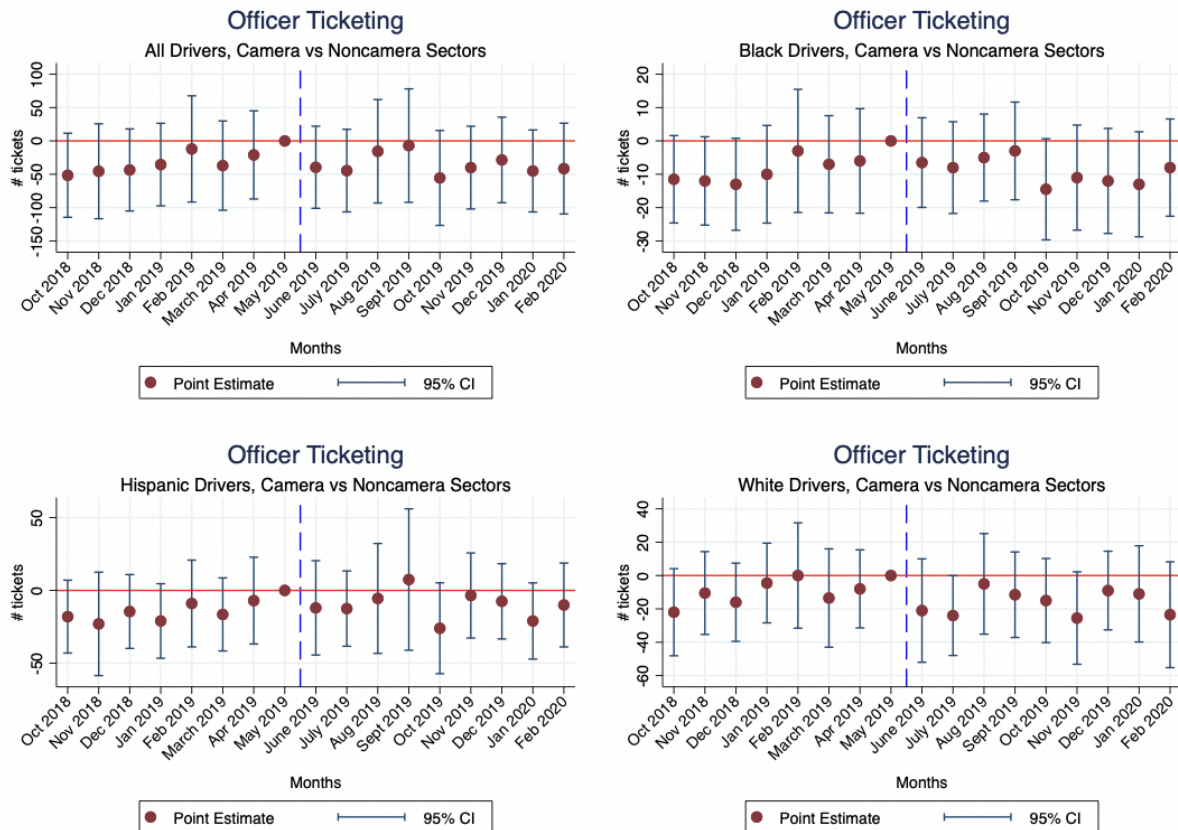
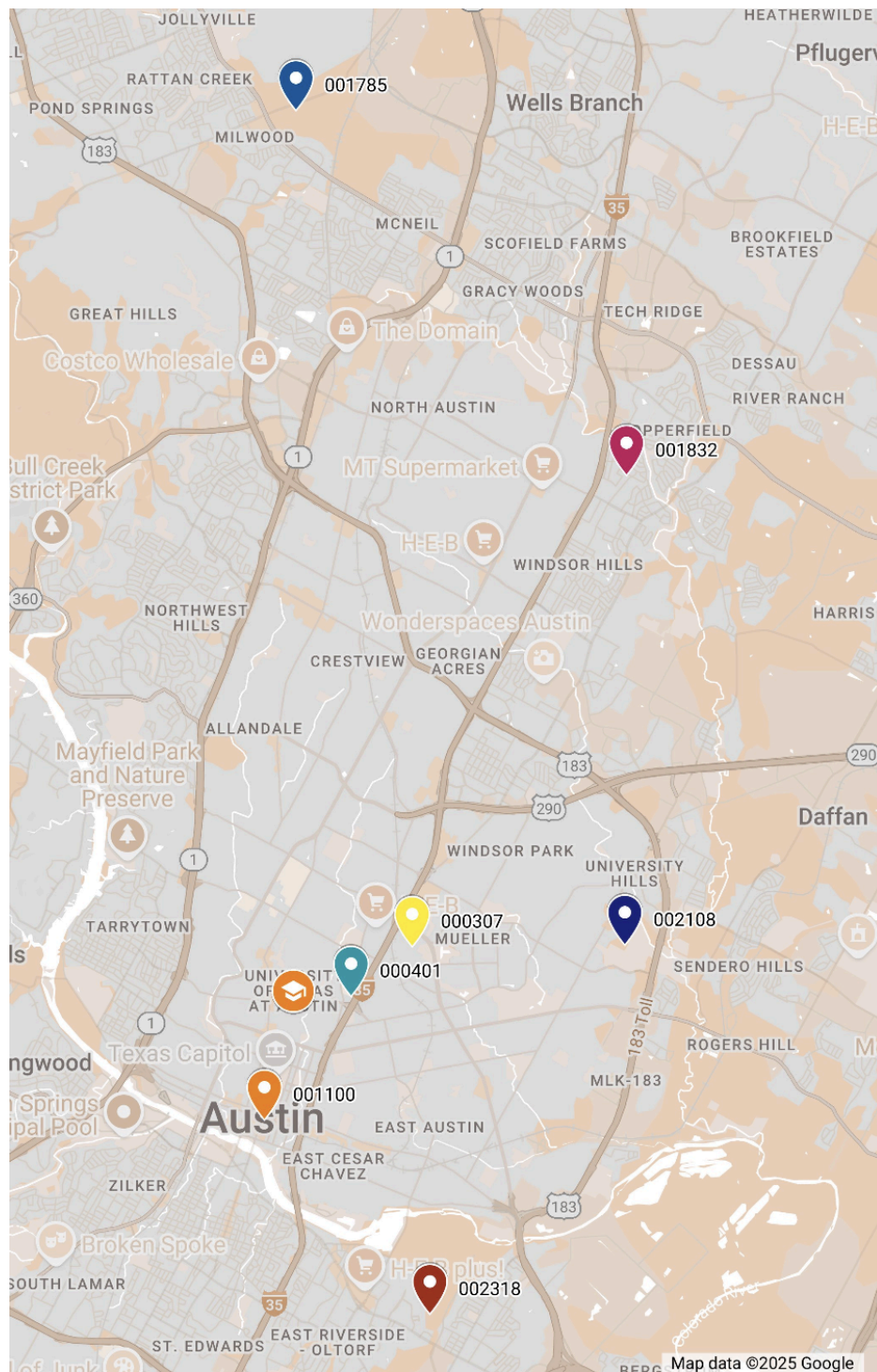


Figure 3. Event study difference in means of officer distributed red light tickets in camera vs non camera sectors overall and for each race



Note: Event studies represent a visual aid of the parallel trends assumption of tickets distributed by officers between camera and non camera sectors. This is done overall and by race.

Figure 4. Where are officers sitting?



Note: This serves as a visual representation of the top seven red light ticketed census tracts by officers after the ban. For context, I include the location of the University of Texas at Austin.

Table 1. Total number of tickets distributed by Red Light Cameras and Officers by Race

	Red Light Cameras Pre-ban	Officers Pre-ban	Officers Post-ban	Total
Black	1500	259	270	2029
Hispanic	1663	606	746	3015
White	5452	827	807	7086
Total	8615	1692	1823	12130

Table 2. Effects of Texas' Red Light Camera Ban on Ticket Distribution

VARIABLES	(1) Overall	(2) Black	(3) Hispanic	(4) White
Camera Sector	139.000*** (10.307)	25.125*** (2.352)	47.750*** (5.119)	66.125*** (4.983)
Post	7.056 (8.696)	1.096 (2.549)	6.931 (4.444)	-0.971 (4.744)
Camera Sector * Post	-23.111 (13.554)	-5.792* (2.979)	-6.861 (8.124)	-10.458 (6.974)
Observations	34	34	34	34
R-squared	0.952	0.931	0.872	0.944
Mean	171.6	16.19	37.88	51.69

Note: These results are measured from officer distributed red light tickets before and after the June 2019 ban on red light cameras in Texas. Each column represents the change in number of tickets, overall and by race, received after the ban in Austin Police Department districts (sectors) with cameras, treated group, and sectors without cameras, comparison group. The ATE is represented in the third row, labeled Camera Sector*Post. Results include seasonal time fixed effects. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Table 3. Effects of Texas' Red Light Camera Ban on Ticket Distribution, aggregated tail months

VARIABLES	(1) Overall	(2) Black	(3) Hispanic	(4) White
Camera Sector	185.333** (74.962)	33.500** (13.844)	63.667** (28.800)	88.167** (33.313)
Post	208.267** (94.666)	-5.000 (19.309)	4.683 (41.992)	-20.217 (55.248)
Camera Sector * Post	-81.033 (85.486)	-16.000 (15.314)	-26.867 (32.838)	-38.067 (38.656)
Observations	32	31	32	32
R-squared	0.627	0.568	0.589	0.663
Mean	133.4	21.58	50.50	68.92

Note: These results differ from Table 2 due to aggregating the tail months together, October 2018 - March 2019 and October 2019 - February 2020, and emphasizing the two months prior and after the ban individually. These results are measured from officer distributed red light tickets before and after the June 2019 ban on red light cameras in Texas. Each column represents the change in number of tickets, overall and by each race, received after the ban in Austin Police Department districts (sectors) with cameras, treated group, and sectors without cameras, comparison group. The ATE is represented in the third row, labeled Camera Sector*Post. Results include seasonal time fixed effects. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Table 4. Effects of Texas' Red Light Camera Ban on Ticket Distribution, day vs night

VARIABLES	(1) Overall	(2) Black	(3) Hispanic	(4) White
Camera Sector * Post * Day	-0.472 (18.840)	-0.810 (4.257)	-5.306 (8.975)	6.042 (8.935)
Observations	68	61	68	68
R-squared	0.878	0.850	0.815	0.869
Mean	52.88	9.593	18.94	25.84

Note: These results are measured from officer distributed red light tickets before and after the June 2019 ban on red light cameras in Texas. Each column represents the change in number of tickets, overall and by race, received after the ban in Austin Police Department districts (sectors) with cameras, treated group, and sectors without cameras, comparison group. The triple interaction is represented in the third row, labeled Camera Sector*Post*Day which signifies if after the ban, there exists a difference in camera vs non camera ticketing by officers in the daytime vs night. Results include seasonal time fixed effects. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Table 5. Where are officers sitting?

Race*Post	(1) Black	(2) Hispanic	(3) White
<i>Census Tract 1785</i>	0.094* (0.056)	-0.003 (0.045)	-0.056 (0.044)
<i>Census Tract 1832</i>	0.058 (0.064)	-0.127** (0.051)	0.089* (0.049)
<i>Census Tract 0307</i>	-0.036 (0.044)	0.059 (0.036)	-0.034 (0.035)
<i>Census Tract 1100</i>	0.031 (0.042)	0.045 (0.034)	0.143*** (0.029)
<i>Census Tract 2108</i>	-0.031 (0.028)	0.032 (0.026)	-0.012 (0.024)
<i>Census Tract 2318</i>	-0.029 (0.026)	0.048* (0.028)	-0.031 (0.029)
<i>Census Tract 0401</i>	-0.085* (0.046)	-0.054* (0.032)	0.110*** (0.031)
Observations	1,403	1,403	1,403

Note: I take the top seven census tracts in which the most officer red light tickets were distributed after the ban. Results are measured from a difference-in-differences model where the treated group is the race in each column compared to the other two races before and after the ban. Estimates are interpreted as the percentage point likelihood that said race is to be ticketed compared to the other two races after the ban in a particular census tract. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Appendix

Figure A1. Distribution of tickets by Red Light Cameras and Officers by day and night time

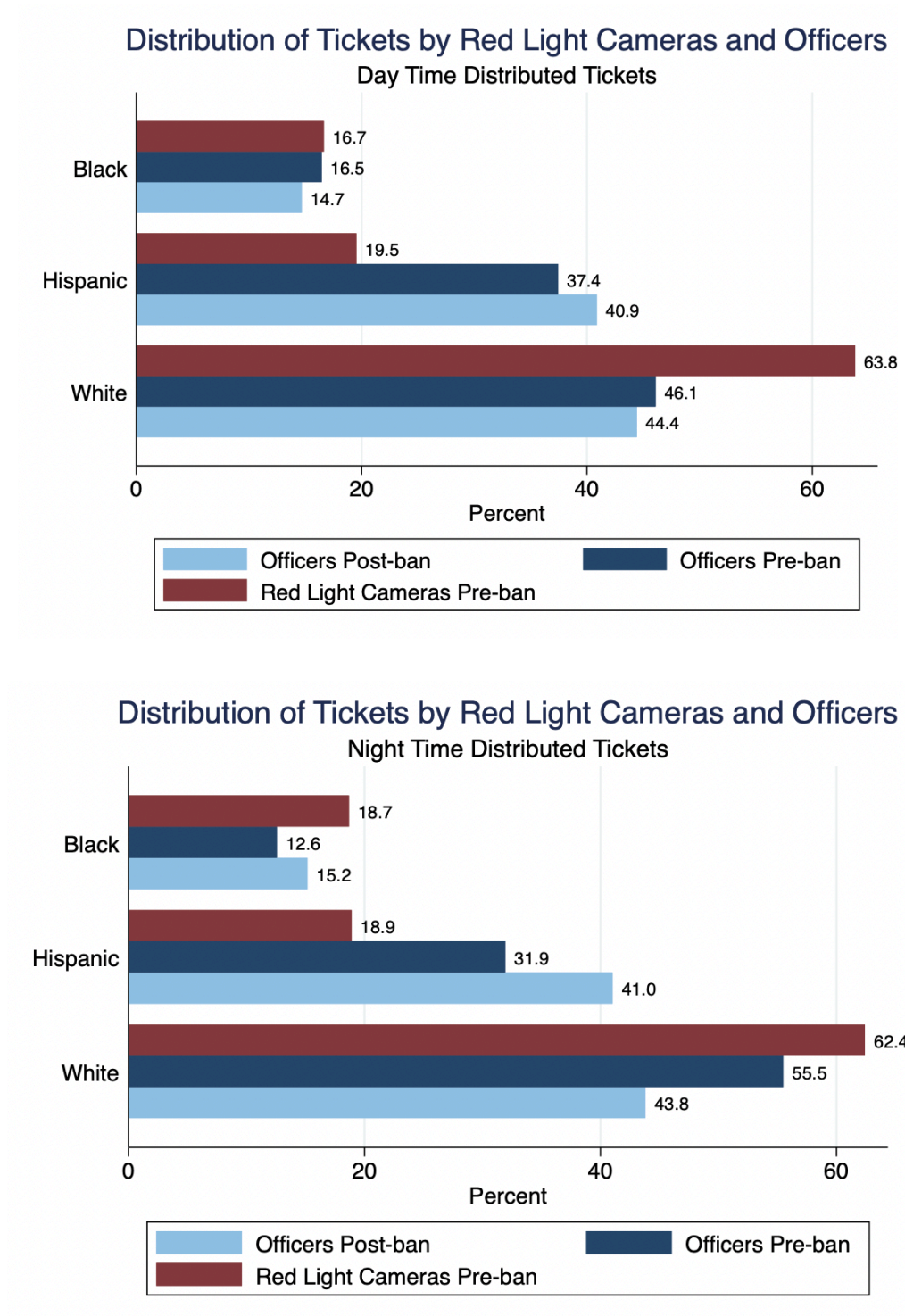


Table A1. Robustness Check: Effects of Texas' Red Light Camera Ban on Ticket Distribution, drop February

VARIABLES	(1) Overall	(5) Black	(6) Hispanic	(7) White
Camera Sector	139.000*** (10.013)	25.125*** (2.373)	47.750*** (5.215)	66.125*** (4.541)
Post	12.313 (10.816)	2.438 (2.512)	7.438 (5.433)	2.438 (5.880)
Camera Sector * Post	-21.125 (13.701)	-6.875** (2.942)	-5.625 (8.842)	-8.625 (6.577)
Observations	32	32	32	32
R-squared	0.957	0.937	0.873	0.956
Mean	171.6	16.19	37.88	51.69

Note: These results differ from Table 2 by excluding February 2020 which makes the pre and post Red Light Camera ban periods both consist of eight months each. These results are measured from officer distributed red light tickets before and after the June 2019 ban on red light cameras in Texas. Each column represents the change in number of tickets, overall and by each race, received after the ban in Austin Police Department districts (sectors) with cameras, treated group, and sectors without cameras, comparison group. The ATE is represented in the third row, labeled Camera Sector*Post. Results include seasonal time fixed effects. Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1